Group Project

Let's Critically Think about IR Applications

GOAL	Collectively reflect about concepts covered in class. Experience all the stages of the research process, from idea-generation to outcome, using Information Retrieval as a case study.
TOTAL POINTS	Up to 10 points (70% of the overall course grade)
HINT	Communication among group members is key. This is your chance to dive deep into an IR topic of your choice (among those offered); you decide if your effort goes into exploration or new solution development. We urge you to think bigger than your grade and instead focus on your learning. Keep in mind that: 1. Many cool and influential ideas, insights, and algorithms started as class projects. 2. Getting the best research-oriented jobs will probably involve giving a job talk. Your project can be the one. 3. You can do a favor to the scientific community by providing data, code, and results (including things that didn't work!).

Overview

One of the most important skills you should develop during your time in graduate school is your ability to be an *independent researcher*. This means that you should focus on your ability to be aware of the area of study you are working on, understand the state-of-the-art (in terms of strategies and methodologies connected to a given research problem), be capable of identifying problems, propose solutions for them (whether that is an empirical exploration or a new technology that can contribute towards advancing knowledge to the field), and design and conduct evaluations to demonstrate the validity of your solution.

Learning Objective

- Develop and put into practice the skills required to be a researcher.
- Demonstrate your ability to juxtapose IR-associated tasks.
- Create and execute a plan that will allow you to independently gain a deeper knowledge of a specific IR-associated task.
- Connect concepts learned in class to areas beyond IR (consider implications of your work/findings).
- Develop teamwork skills.

Details Specifications

For this group project, you will first need to form a **group of 4** individuals and self-assign to a Group on Brightspace.
Together, you will pick an IR area of interest and then decide to either undertake an *empirical* analysis of methods/solutions, develop *new* methods/solutions, or undertake a *reproducibility* study—this will

determine the expected main contribution of your work. You will have a month to work on this project, produce a report, and discuss outcomes with class instructors.

As you choose a topic and define your approach to addressing a problem, keep the following in mind:

- Empirical analysis of existing methods or solutions: Your main contribution should come from the depth and quality of your analysis. This includes thoroughly comparing and contrasting results across different settings, strategies, or configurations.
- Development of new methods or solutions: Your contribution should focus on clearly motivating and justifying your design decisions, and on convincingly demonstrating the validity and effectiveness of your proposed method or solution.
- Reproducibility studies: Your work should aim to repeat, reproduce, generalize, or analyze prior research relevant to information retrieval. The emphasis is on producing new insights or findings about established approaches.

Topics. We encourage you to make full use of the notebooks provided for this course (https://github.com/wis-delft/ms-information-retrieval), along with the insights you gained from exploring them. Drawing inspiration from current research in areas such as indexing, query processing, relevance estimation, ranking models, recommender systems, conversational search, and related topics, your team should identify a research gap worth investigating.

Because the goal is for you to define your own research direction, we are not prescribing specific topics. Instead, we ask you to discuss, as a team, where you believe meaningful contributions can be made and how you plan to pursue them. As a starting point, we encourage you to focus on a research gap for which you can leverage the existing notebooks or literature that already provides public code repositories. This will give you a groundwork implementation foundation—especially important given the limited timeline for your project.

To ensure you start your project on a strong foundation, focus your literature review on reputable venues such as the ECIR, SIGIR, CHIIR, RecSys, UMAP, CIKM, ICTIR, and WSDM conferences, as well as the TOIS, TORS, and TWeb journals.

Assessment. This is a group assignment.

We will not judge your project based on how "good" the results are!

In the real world, publication venues do this, because they have extra limits on space that make them prefer positive evidence for new developments over negative results. In this course, we care about your learning experience and therefore will value positive results, negative results, and everything in between. Consequently, we will judge your project based on:

- Soundness of the methodology.
    - Setup of empirical analysis (data, metric, procedure)
    - Design choices of new method
- How well the metrics for experiments fit.
- Insightfulness of the results.

  - Results interpretation
  - Discussion on implication
  - Limitation of findings
  - Implications of findings to the research community (technical novelty) and society at large (broader impact)

Your final grade for this project will be distributed as follows:

- Quality of content (35%)
  - Technical correctness; depth, results interpretation; discussion.
- Interview (30%)
  - Understanding; insights.
- Report writing (30%)
  - Details, coherence, structure, and use of English.
  - Related work, as in discussion presented in related work section focused on existing literature that informs and contextualizes your proposed work.
- Peer assessment (5%)
  - Contribution to the project

Deliverable & Deadlines. You are encouraged to submit on *Brightspace* a brief outline of your idea. This will not be graded but instead will be an opportunity for formative assessment (and guidance, if needed, on managing scope and/or experimental set up).

If you are proposing your own topic (i.e., going beyond the notebooks we shared for foundation), then you must submit this initial outline so that we can sign off on it.

By the end of Q3, you will need to submit on *Brightspace* the final project report, you will also need to include a link on your report to the repository when you have the code for this project.

Submissions should be 1 per group, in PDF, using the required template. After submission, a plagiarism check is executed by Turnitin—please make sure to follow the academic rules of writing (you can read them up here one more time).

At the end of the project, we will conduct an interview about the group project and its connection to the course content. Based on the project report and the interview, each member of the group receives a grade. This interview will last between 15 to 25 minutes per group.

- *Brief outline of proposed work* (required only for self-proposed topics): 23:59 CET, March 7th, 2025
- *Final report*: 23:59 CET, April 3th, 2025. This is mandatory and graded.
- *Interviews* (in person): The week of April 10th, 2025. We will schedule a specific day/time per group at least two weeks prior.

Template for reports. The **final** group project report should be **4 pages** long in Association for Computing Machinery (ACM) - Generic Conference Template (2 columns). A 'References' section is required but excluded from this page limitation. Also excluded from this page limitation are your discussion on 'Self-Assessment of Contributions', 'Link to Repository', and 'Declaration of Generative AI Usage'.

Please keep in mind that all main content—including figures and tables that can help you illustrate some points and give a bit of breathing room to all the text—*must fit within the 4-page limit*. This is the material we will use for grading, as your contribution must be self-contained.

We will share additional tips during the lecture when we introduce the group project. In essence, focus your narrative on the **research aspects**, not the **implementation details** 😉

The report should include the following:

- ***Title, group id, authors.***
- ***Abstract***. Ideally a paragraph long.
  - It's good to give some background for the work right at the beginning, define the current proposal and situate it in that background, summarize the main findings, and end by identifying the wider significance of the work. The "General reasoning" section of your experiment protocol is likely to provide good content for the abstract. Remember that an Abstract is not a summarized Introduction. Instead, it is a higher-level description of your work.
- ***Introduction***. Problem statement, overall plan to tackle the problem.
  - This is a particularly important part of the paper. In this section, the reader is likely to form their expectations for the work and start to form their opinions on it. The introduction should tell the whole story of the paper, in a way that is understandable to most people in the field:
    - Where are we? That is, what area of the field are you working in? Answering this question is important for orienting the reader.
    - What is the goal of the work presented in this paper? If it is a new method/solution, that often includes a hypothesis. If it is an empirical analysis, that instead generally refers to the need for comparing/contrasting.
    - What concepts does your work rely on? You can't expect your reader to fill in the gaps. When necessary, define terminology as part of your narrative.
    - What are the main findings of the paper?
  - By the end of the Introduction, the reader should be able to know what the problem of interest is, what makes it a problem, what did you propose to do and why, and what are the main contributions/lessons learned emerging from this work.
- ***Related Work***. Background and closely-related literature informing and/or giving grounding to your work.
  - A good strategy for this section is to first group the papers you want to cover into general categories that relate to your own work in important ways. For each such category, state its thematic unity, briefly discuss what each paper accomplishes, and then, importantly, relate this work to your own project, as a way of providing background for your work and distinguishing it from previous work. In this way, you create a space for your own contribution.

  - We encourage you to avoid summarizing individual papers, and instead salient take aways from subsets of papers related to your work. That way you showcase your understanding of the foundational knowledge and gaps inspiring your work, while being mindful of page limitations.
- ***Method***[1]. Length highly variable.
  - If you are discussing an empirical analysis, then this is the section where you describe the framework for your analysis, including, but not limited to, data, metrics, procedures, and methods/solutions under analysis.
  - If you are discussing a new method/solution, then this is the section where you describe the technical details and design decisions of your proposed method/solution.
- ***Experiments***. Length highly variable.
  - If you are discussing an empirical analysis, in this section you provide an in-depth discussion of experiments conducted, results, and inferences that can be made from findings.
  - If you are discussing a new method/solution, here is where you discuss experiments (including data, metrics, and baselines for comparison) and results that demonstrate correctness/effectiveness/efficiency of your proposed method/solution.
  - Regardless of the focus of your work, in this section you should also discuss limitations.
  - Discussion of implications and broader impacts of your work and connection to literature.
- ***Concluding Remarks***. Describe what you learned/found and what avenues for further improvement you see.
  - It's best to briefly summarize what the paper did and why and then try to articulate the broader significance of the work, looking ahead to expanding its scope.
  - In papers of 4 pages in double column, concluding remarks tend to be very short, i.e., a direct high-level reiteration of the point and valuable outcomes of the work presented.
- ***Self-Assessment of Contributions*** [excluded from page limitations]. Each student should briefly outline their contribution to this project.
  - Here you can discuss implementation responsibilities, writing, data collection, management, etc.
  - This section does not count towards page limitations.
- ***Link to Repository*** [excluded from page limitations]**.** You should include a link to a repository that contains the software you created, the scripts you used to run experiments and/or analyze your data, datasets you created, if any.
- ***Declaration on Generative AI*** [excluded from page limitations]. Inspired by CEUR-WS Policy on AI-Assisting Tools, we ask that you include a brief statement on usage of generative AI, if any.
  - Specifically, we ask that you disclose Tools and Services used and Tools' Contributions (here is where you could mention prompts used).
  - The Taxonomy of GenAI Usage might be useful: GenAI Usage Taxonomy

[1] Note that depending on the type of research gap you address, Method and Experiments, could be replaced with Experimental Methodology (Set Up) and Results & Discussion.

  - Although you are allowed to use generative AI tools as writing aids and for brainstorming or clarifying concepts (while keeping in mind their limitations, such as hallucinations), we expect the original thinking and intellectual work to be entirely your own.
- ***References*** [excluded from page limitations].
  - We encourage you to include complete citations in a consistent format. For this, using the bib reference files from Google Scholar or dblp: computer science bibliography would help.
  - As we have discussed during the lecture, papers posted on Arxiv while offering easy access are not always peer-reviewed. With that in mind, when citing papers you found on Arxiv, always include their peer-reviewed citation version.

Expectation for the Interviews. The interview will be an academic discussion about the executed project in the context of this course. The 15–25-minute interviews (time depends on how many groups we have in the end) per group will be scheduled **across the week of April 10.** They will cover the report and connections of the project work to the core IR lectures.